

Marco Mo

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SUMMARY

AI/ML Engineer building agentic systems, document retrieval and fraud models. Delivers tested APIs and staff applications from ingestion through evaluation, serving and monitoring, with privacy controls, human review and operational dashboards.

EXPERIENCE

Process Optimization, Automation & AI Specialist Aug 2026 – Present
Multi-Post

- Consolidated twelve preparation, reporting and audit workflows in a staff-used Django application.
- Reduced selected artwork preparation from approximately 30 minutes or more to seconds.
- Built validated production dashboards and machine-event reports from requirements.

Junior AI Specialist Jun 2026 – Aug 2026
ADC Nederland

- Developed four systems for order ingestion, MCP access, agentic chat and process mining.
- Built an MCP service exposing 67 structured tools over REST and MySQL.
- Combined typed APIs, tenant-scoped authentication, role permissions, human review and tests.

Teaching Assistant Sep 2021 – Jul 2025
Eindhoven University of Technology

- Coached 80+ students weekly in Python, SQL, machine learning and analytical reasoning.

PROJECTS

Agentic Wihabo Database Chatbot with Role-Based Access

- Routed LangGraph requests through deterministic responses, MCP tools, planning and refusal paths.
- Enforced role-based permissions; grounded answers in tool results and screened unsafe requests.
- Reduced LLM calls through deterministic logic and configurable model routing.
- Integrated FastAPI, Streamlit, checkpoints, evaluation runners and automated tests.

Fraud Detection MLOps Pipeline — IEEE-CIS

- Built training and threshold optimization for approximately 600,000 transactions and 400+ features.
- Prevented temporal leakage through chronological splits and held-out streaming evaluation.
- Achieved 0.9287 ROC-AUC with XGBoost; served the model through FastAPI.
- Tracked MLflow experiments and feature drift; automated Docker builds with GitHub Actions.

Offline RAG System for Legal Contracts

- Kept indexing, retrieval, reranking and generation local across 400+ legal documents.
- Combined BM25, dense retrieval, reciprocal-rank fusion and cross-encoder reranking in Qdrant.
- Constrained Ollama generation to retrieved context and displayed supporting sources in Streamlit.

EDUCATION

MSc Data Science and Artificial Intelligence 2023 – 2025
Eindhoven University of Technology

- Thesis: Reward-Free Safe Reinforcement Learning Exploration Using Different Entropy Measures

BSc Data Science 2020 – 2023
Eindhoven University of Technology and Tilburg University

- Bachelor End Project: Health Platform Text Classification Using Active Learning

SKILLS

AI/ML: scikit-learn, XGBoost, PyTorch, TensorFlow, NLP, time-series analysis, reinforcement learning

LLM/RAG: LangGraph, LangChain, MCP, embeddings, Qdrant, Ollama, guardrails

Engineering: Python, FastAPI, Pydantic, MLflow, model serving, monitoring, Docker, CI/CD, GitHub Actions

Data: SQL, pandas, NumPy, ETL, schema validation, PostgreSQL, MySQL, Parquet

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)